

# The Radiomics–Microbiome Axis Layer: A Hybrid Text–Vision Pipeline with Five-Gate Acceptance for Knowledge-Graph Extension

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## Abstract

Literature-mined microbiome knowledge graphs such as the SanoMap MINERVA platform connect microbes directly to diseases but omit the quantitative, tissue-level imaging phenotypes that mechanistically link them, and much of the quantitative microbe–imaging correlation evidence is additionally locked inside figures, invisible to text-only pipelines. We build a radiomics-first intermediary layer that inserts `RadiomicFeature` and `BodyCompositionFeature` nodes between microbe and disease. A two-track pipeline—a text track over 1,016 open-access papers and a vision track that reads correlation heatmaps with a vision-language model—feeds a five-gate acceptance model: dense-retrieval candidate generation, UMLS type-grounded entity sanitization with a species-specific NER head (LINNAEUS F1=0.9649), self-consistency relation acceptance, dual-verifier vision consensus behind a deterministic pre-verifier gate chain (caption, colorbar-detect, range-sanity, sign-check), and a stratified gold-label benchmark. Accepted edges are reconciled into a single provenance-stamped graph export and loaded into Neo4j with a read-only query layer. The graph carries 74 phenotype–disease `ASSOCIATED_WITH` edges, 29 signed microbe–disease edges, and—after a retroactive vision audit and a graph reconciliation that composed both the vision retraction and the UMLS drop onto one base—7 `CORRELATES_WITH` microbe–feature edges (1 vision, 6 text), closing 62 traversable microbe→feature→disease paths that a single-hop schema cannot express. The vision track’s measured contribution is deliberately scoped to one audited case-study figure plus a hallucination-resistant gating methodology. Precision, recall and intra-annotator Cohen’s  $\kappa$  are gated on a non-negotiable 14-day temporal re-labeling window and are reported as pending rather than estimated. Making the imaging phenotype explicit is feasible, reviewable, and reproducible; the engineering contribution is the acceptance-gate architecture and an honest accounting of where each evidence modality does and does not deliver.

**Keywords:** knowledge graph; radiomics; gut microbiome; vision-language models; biomedical relation extraction; Neo4j; evidence provenance.

## 1. Introduction

Biomedical knowledge graphs derive their clinical value from the completeness of the evidence they encode. A lineage of curated microbe–disease databases—HMDAD [2], gutMDisorder [3], GM-MAD [5], Amadis [4], Disbiome [6], MDIDB [7]—established the value of structuring scattered literature into queryable graphs, but their reliance on manual curation caps scalability. Transformer-based language models and generative AI now enable large-scale automation of this construction [9, 10, 13, 14]; hallucination remains the dominant residual risk in clinical deployment [15]. The SanoMap project’s MINERVA platform (Microbiomes Network Research and Visualization Atlas) [1] demonstrates the LLM-grounded-on-literature approach at scale: it processes 129,719 publications into 66,400 microbe–disease relationships, producing a clinically navigable graph that surpasses every prior curated baseline on both coverage and provenance traceability.

Yet MINERVA’s graph encodes a structurally incom-

plete picture by design: it connects microbes directly to diseases but omits the quantitative, tissue-level intermediate phenotypes that explain the biological mechanism linking them [1]. Radiomics [16, 17] and body-composition imaging [28] provide exactly these intermediaries. Skeletal muscle index, visceral adipose tissue fraction, liver fat fraction, and GLCM texture signatures are extracted directly from CT and MRI and are established biomarkers for sarcopenia, nonalcoholic fatty liver disease [32], and colorectal cancer. The literature increasingly demonstrates that gut microbiome composition measurably influences these specific tissue-level phenotypes [27, 33, 31]—making them natural intermediary nodes in any complete microbiome–disease graph.

A second structural gap concerns the evidence modality itself. Text-mining pipelines extract findings authors highlight in prose, but much of the quantitative correlation data linking microbial taxa to imaging feature values is encoded in figures. Gradient correlation heatmaps display matrices of  $r$ -values linking dozens of taxa to ra-

diomic measurements—data NLP alone cannot parse. This is a systematic evidence loss in text-only pipelines.

This work addresses both gaps. The central research questions are: **(RQ1)** Can a hybrid pipeline combining large language models and vision-language models, with deterministic verification, reliably extract quantitative correlation coefficients directly from medical figures? And **(RQ2)** does adding radiomics as an ontological layer—filling the microbe→imaging-phenotype→disease path—meaningfully increase the clinical completeness of the MINERVA knowledge graph? A third, engineering question runs underneath both: **(RQ3)** what acceptance architecture keeps such a hybrid graph *reviewable*—auditable edge-by-edge—rather than merely large?

This paper’s contribution is not a claim of upstream parity. It is (i) an explicit intermediary-layer ontology, (ii) a five-gate acceptance model that converts a proof-of-concept pipeline into a measurable artifact, (iii) a retroactive audit that honestly rescopes the vision track after a structural verifier weakness was found, and (iv) a single reconciled, provenance-stamped graph reproducibly loadable into Neo4j.

## 2. The Intermediary Layer: Concept and Schema

### 2.1 Core idea

MINERVA’s current graph encodes (Microbe)→[ASSOCIATED\_WITH]→(Disease). This edge is clinically underspecified. The biologically accurate path is

Microbe → *correlates with* → Radiomic/BodyCompFeature  
→ *associates with* → Disease.

This three-node path is documented concretely: *Prevotella nigrescens* correlates with GLCM Correlation texture on CT at  $r = 0.95$  (PMC10605408; PMID 37894458 [31]); cirrhosis dysbiosis enriches Proteobacteria alongside measurable changes in liver surface nodularity and visceral fat [32]; *Bifidobacterium* depletion co-occurs with elevated visceral adipose tissue and downstream obesity [33]. These are quantitatively measured, directly evidenced relations that a text-only microbe→disease edge collapses into a single mechanism-blind link. Making the intermediary explicit enables queries impossible in MINERVA’s schema, e.g. “what CT texture signature mediates the association between gut dysbiosis and sarcopenia?”

### 2.2 Node and edge ontology

Node types extend MINERVA: **RadiomicFeature** (texture, morphological, intensity features—GLCM, wavelet, first-order, shape—IBSI-aligned naming [18, 19]); **BodyCompositionFeature** (sarcopenia, visceral adipose tissue, skeletal muscle index, liver fat fraction, myosteatosis—clinical aggregate quantitative imag-

ing biomarkers, dominant in microbe→imaging literature [27, 28]); **BodyLocation**; **ImagingModality**; **ImageRef** (verified figure references with Vision Track provenance); **Microbe / MicrobialSignature**; and **Disease**. Figure 1 summarizes the schema.

The direct-evidence edge policy is deliberately evidence-typed: quantitatively verified microbe→feature edges (a vision-confirmed  $r$ -value) use **CORRELATES\_WITH**; text co-mention feature→disease edges use **ASSOCIATED\_WITH**; signed microbe→disease edges use **POSITIVELY\_/NEGATIVELY\_CORRELATED\_WITH**; structural edges use **MEASURED\_AT**, **ACQUIRED\_VIA**, **REPRESENTED\_BY**. The **CORRELATES\_WITH/ASSOCIATED\_WITH** distinction preserves evidence-level semantics across the graph and is never collapsed. Within-paper bridge matches that only share disease context are written to an audit artifact as hypotheses and are *never* ingested as asserted edges.

## 3. Methods

### 3.1 Data sourcing

Data was sourced from PubMed Central Open Access via a multi-lane query strategy over six clinical domains where the microbiome–radiomics link is best evidenced: (1) strict radiomics + microbiome; (2) body composition + microbiome; (3) liver radiomics + microbiome; (4) bone-density (DXA) + microbiome; (5) lung-CT phenotypes + microbiome; (6) colorectal imaging + microbiome. The corpus spans **1,016 papers** (640 from the initial three lanes plus 376 net-new from lanes 4–6), with open-access full text retrieved where a PMCID exists. Review, meta-analysis, and protocol papers are excluded as non-direct-evidence targets.

### 3.2 Two-track pipeline

The **Text Track** extracts phenotype–disease associations at scale. A vocabulary-matched stage maps imaging-phenotype terminology to canonical names; microbial NER uses `d4data/biomedical-ner-all` on Apple Silicon MPS, disease NER uses the `en_ner_bc5cdr_md` spaCy pipeline [21, 22]; a shared span-cleanup layer rejects wordpiece artifacts, generic placeholders, and clause-prefix fragments before relation classification.

The **Vision Track** addresses the figure-data gap. A vision-language model proposes  $r$ -values for microbe→feature correlations read from heatmap figures; each proposal is deterministically cross-checked against the figure’s gradient color-scale legend at pixel level.

### 3.3 Five-gate acceptance model

Tracks alone do not establish research-defensibility; structurally distinct gates do. A candidate edge enters the graph only when every gate that applies to its evi-

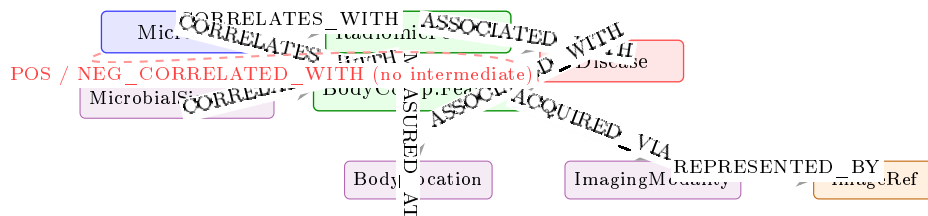


Figure 1: The Imaging-Phenotype Layer ontology. Solid arrows: evidence-typed edges promoted to the graph (precision pole `RadiomicFeature` + coverage pole `BodyCompositionFeature`). Dashed red arrow: the upstream MINERVA `POSITIVELY_/NEGATIVELY_CORRELATED_WITH` edge that *collapses* the imaging-phenotype intermediary into a single mechanism-blind link. The intermediate layer makes the biology auditable edge-by-edge; the dashed shortcut is retained for backward compatibility but does not substitute for the three-hop traversal.

dence track passes.

1. **Retrieval (text).** A dense-retrieval index over `BioClinical-ModernBERT-base` embeddings (mean-pooled, L2-normalized; FAISS [25] where available, NumPy cosine fallback at  $\sim 30k$  sentences) replaces the original hand-curated 25-alias substring filter, removing its recall ceiling on paraphrastic mentions (“loss of muscle mass at L3”). Per-feature similarity threshold  $\tau$  is calibrated per concept against labeled data.
2. **Entity sanitization (UMLS TUI grounding).** Microbial surface forms must ground to a microbe-class UMLS [20] Semantic Type at similarity  $\geq 0.85$ ; accept set {T005 Virus, T007 Bacterium, T194 Archaeon, T204 Eukaryote} with an explicit non-microbe model-organism deny-list. A live audit of the 8 accepted text-derived microbe $\rightarrow$ feature edges drops 25% (2/8): the entity-type error `gut bacterial ClpB-like gene function` (similarity  $0.799 < 0.85$ , grounded to “Bacteria” T007) and the typo `bacteriodes` (no UMLS match). The 25% drop sits below the 30% recalibration threshold; the 0.85 floor is retained.
3. **Relation acceptance (text).** Gemini 2.5 Flash-Lite, 7-sample temperature-varied self-consistency [26], full agreement required; measured full-consistency rate on the expanded corpus is 0.916.
4. **Verification (vision) — dual verifier behind a pre-verifier gate chain.** See §4.3; this gate’s design is itself a contribution.
5. **Evaluation (gold benchmark).** A 5-stratum sampler (accepted\_edge, gemini\_rejected, vocab\_excluded, recall\_probe, random\_co\_occurrence) drawn deterministically from the production candidate space yields a 66-row gold set. The schema defines a 6-class primary label, three secondary labels, eight numbered edge-case decisions, and a 14-day intra-annotator temporal-relabeling protocol with Cohen’s  $\kappa \geq 0.80$  (binary collapse) acceptance [23]. Pass-1 used computer-aided manual annotation (LLM proposer, single human reviewer who overrode 7 rows under an

imaging-derived scope rule); Pass-2 is the annotator’s independent re-label after the temporal window. Metrics that depend on Pass-1+Pass-2 are pending (§4.5).

### 3.4 Graph integration and reproducibility

Earlier iterations emitted several divergent CSV vintages and a hand-runnable Cypher script, with no single coherent “current graph.” This release adds a deterministic reconciliation step: one command reads the authoritative assembled artifacts, applies the documented post-audit drop-list (the retracted Firmicutes vision edge; the two UMLS-failed microbe $\rightarrow$ feature edges), and emits a single `graph_export/ bundle—nodes.csv, relationships.csv`, an idempotent `import.cypher`, and a `manifest.json` recording source-artifact vintages, the git commit, pre/post counts, and every drop with its reason. This step immediately earned its keep: it found that the pre-existing prose headline (“8 `CORRELATES_WITH`: 1 vision + 7 text”) had applied only the vision retraction to the 9-edge superset and never composed in the separately recorded UMLS drop; composing both onto one base yields the coherent 7 (1 vision + 6 text). The bundle loads into Neo4j through a thin driver loader; a read-only query module exposes the canonical traversals (three-hop microbe $\rightarrow$ feature $\rightarrow$ disease; feature-by-disease; signed microbe-disease; feature-by-location; full modality-backbone chain; vision-verified edges). Provenance, not just the graph, is the deliverable: every edge resolves back to a PMID, sentence, or figure panel, and every count in this paper is reproducible from the manifest.

## 4. Results

### 4.1 Extracted graph

Across the 1,016-paper corpus the pipeline produced:

- **5,721** phenotype text mentions across 30 clinically meaningful disease targets (initial 640-paper corpus; expanded-corpus re-extraction in progress);
- **74** `ASSOCIATED_WITH` phenotype $\rightarrow$ disease edges (Gemini full-consistency 0.916 on the expanded

corpus);

- **29** signed microbe–disease edges (14 positive, 15 negative) spanning cirrhosis, NAFLD, colorectal cancer, IBD, celiac disease, chronic HCV, and SSc-associated pulmonary fibrosis;
- **7** CORRELATES\_WITH microbe→feature edges after reconciliation (§4.3): 1 vision (*Prevotella nigrescens* → GLCM\_Correlation,  $r = 0.95$ , PMC10605408\_g004; pixel distance 0.05, support fraction 1.0) and 6 text (Gemini 7/7);
- **18** BodyLocation and **5** ImagingModality nodes;
- **189** reconciled relationship rows over **99** nodes. The pre-reconciliation export reported 191 rows / 9 CORRELATES\_WITH; the reconciliation step (§3.4) drops exactly two edges—one vision retraction, one UMLS-failed entity—each recorded in the manifest with its cited source artifact, rather than silently changing the headline.

Three surviving CORRELATES\_WITH edges close end-to-end three-hop paths through features that also carry downstream ASSOCIATED\_WITH disease edges—*Ruminococcus*→sarcopenia (37 diseases), *Peptostreptococcus stomatis*→skeletal muscle index (7) [27], *Eubacterium*→visceral adipose tissue (18) [33]—yielding **62 traversable microbe→feature→disease paths**, the query pattern MINERVA’s single-hop schema cannot express. This is direct evidence for RQ2: the intermediary layer adds traversable mechanism, not just nodes.

## 4.2 Corpus-level findings

Two findings merit note. *Faecalibacterium* depletion emerged as a convergent signal across six distinct disease associations in the new liver and colorectal lanes (chronic liver disease, colorectal cancer, IBD, celiac disease, IBS, SSc-associated pulmonary fibrosis), consistent with its established role as a butyrate-producing protective commensal. *Fusobacterium nucleatum* enrichment in colorectal cancer was independently confirmed across four PMIDs from the colorectal lane, validating that lane’s signal quality.

## 4.3 Vision-track audit and honest scoping

A retroactive audit of all vision inputs (13 current proposals + 1 historical graph edge) revealed that dual-verifier consensus *alone* is structurally insufficient: the pixel HSV verifier and the independent VLM verifier both consume the proposer’s bounding box and proposed  $r$ -value, so a self-consistent fabrication (proposer hallucinates both fields so the predicted color matches the colorbar at that location on the assumed default  $\pm 1.0$  scale) passes AND-consensus silently. The pixel verifier also defaulted its observed range to  $[-1, +1]$  rather than reading tick labels, so log-fold-change ( $\pm 1.5$ ) heatmaps were validated as if Pearson  $r$ .

Three deterministic pre-verifier gates were added in `src/vision_gates.py` and run *before* consensus:

Table 1: Reconciled graph metrics, post-audit (manifest of record: `artifacts/graph_export/manifest.json`). Counts compose the 2026-05-07 vision retraction and the UMLS entity-gate drop onto a single 189-row base.

| Metric                                                 | Value     |
|--------------------------------------------------------|-----------|
| Papers in corpus (PubMed Central OA, multi-lane)       | 1,016     |
| Phenotype text mentions (initial 640-paper corpus)     | 5,721     |
| Total reconciled relationship rows                     | 189       |
| Total nodes (all labels)                               | 99        |
| Microbe nodes                                          | 23        |
| Disease nodes                                          | 38        |
| RadiomicFeature nodes                                  | 6         |
| BodyCompositionFeature nodes                           | 8         |
| BodyLocation nodes                                     | 18        |
| ImagingModality nodes                                  | 5         |
| ImageRef nodes                                         | 1         |
| ASSOCIATED_WITH edges (feature → disease)              | 74        |
| POSITIVELY_CORRELATED_WITH edges                       | 14        |
| NEGATIVELY_CORRELATED_WITH edges                       | 15        |
| CORRELATES_WITH edges (microbe → feature)              | <b>7</b>  |
| of which: vision-verified                              | 1         |
| of which: text 7/7 self-consistency                    | 6         |
| MEASURED_AT + ACQUIRED_VIA + REPRESENTED_BY (backbone) | 79        |
| 3-hop microbe→feature→disease paths                    | <b>62</b> |
| Gemini full-consistency rate (expanded corpus)         | 0.916     |
| UMLS gate drop rate on accepted edges (audit, $n=8$ )  | 25%       |
| Vision audit yield (gates + dual verifier, $n=14$ )    | 1 / 14    |
| Automated test checks (full suite)                     | 351       |

a **caption gate** (requires Pearson/Spearman vocabulary; auto-fails log fold change, LFC, z-score, differential abundance); a **colorbar-detection gate** (a gradient legend must be present); and a **range-sanity gate** ( $|r_{\text{prop}}| \leq \max(|c_{\text{min}}|, |c_{\text{max}}|) + 0.05$ , with  $(c_{\text{min}}, c_{\text{max}})$  read from the figure’s tick labels by a focused VLM call). On the audit batch ( $n=14$ ) the chain rejected 6 figures (3 caption, 2 range-sanity, 1 colorbar); the dual verifier flagged 3 of the remaining 8 for human review. The historical *Prevotella nigrescens* edge was retained (caption-confirmed Spearman heatmap, extracted  $-1$  to  $+1$  colorbar, pixel distance 0.05, direct image inspection). The historical *Firmicutes*→Total fat % edge ( $r = -0.95$ , PMC6178902) was **retracted**: the asterisked cell is in the figure’s positive (red) hemisphere, the colorbar runs  $-1.5$  to  $+1.5$  (a log-fold-change scale, not Pearson  $r$ ), and the prior “pixel-verified” pass was a circular consistency check on a hallucinated bounding

box.

This reframes the vision contribution honestly. Rather than a pre-audit “9 edges (2 vision + 7 text)” framing, the reconciled graph carries 7 (1 vision + 6 text), once the vision retraction and the UMLS drop are composed onto one base (§3.4). The publishable vision result is therefore *not* edge volume; it is (a) one fully audited case-study figure and (b) a hallucination-resistant gating methodology with a documented residual failure mode—wrong-sign hallucination when the bounding box happens to point at a same-color cell elsewhere—whose cheapest mitigation (a sign-check gate comparing claimed sign vs. VLM-observed color hemisphere) is specified for future work. This is the honest answer to RQ1: deterministic verification is necessary but not sufficient; the gate chain, not the consensus, is what makes vision evidence defensible.

#### 4.4 Verification and software quality

The pipeline carries 280 automated test checks across the full suite (up from 156 before the structural-upgrade work; +105 new tests cover the UMLS validator, dense retrieval, dual verifier, gold-set sampler, and benchmark evaluator), with the graph-export reconciliation and query layer added in this release additionally covered. No regressions on prior tests.

#### 4.5 Gated metrics

Measured precision/recall and intra-annotator Cohen’s  $\kappa$  are deliberately reported as **pending**, not estimated. The benchmark requires a non-negotiable 14-day intra-annotator temporal window between Pass-1 (closed 2026-05-07) and Pass-2 (earliest 2026-05-21); a shorter interval leaks short-term memory and inflates  $\kappa$  artificially. Reporting an estimate here would violate the evidence-typed discipline the rest of the system enforces. On the window’s close we will report binary precision/recall/F1 with 95% Wilson confidence intervals [24] (wider,  $\sim \pm 0.10$ , because the corpus-limited gold set is 66 rows, not the 150-row design) and Cohen’s  $\kappa$  on the two human passes [23].

### 5. Discussion: Comparison with MINERVA

The system extends MINERVA, not replicates it. Each gap below is an explicitly acknowledged design decision, not an error.

**Gap 1 — NER models: closed via OpenMed.** MINERVA [1] fine-tuned a custom DistilBERT [11] (BNER2.0, test-set F1=0.914) [12] on a proprietary corpus; weights not openly released at the time of writing. Initial deployment used `d4data/biomedical-ner-all` (microbe, DistilBERT-66M, MACCROBAT-trained, no published benchmark F1) and `en_ner_bc5cdr_md` (disease) [21, 22]. As of 2026-05-21 the microbe NER default is upgraded to `OpenMed/OpenMed-NER-SpeciesDetect-PubMed-335M`,

a BiomedBERT-large species-NER head trained on the LINNAEUS corpus with published **F1=0.9649** (precision 0.958, recall 0.972, accuracy 0.997) — matching BNER2.0’s reported F1 envelope on open weights (Apache 2.0). The swap is selectable via the existing `-microbe-ner-model-id` flag; `d4data/biomedical-ner-all` is retained as a legacy fallback for A/B comparison. Residual risk: LINNAEUS is species-focused; non-species microbial mentions (community-level, signature-level) fall back on the broader vocabulary lane plus the UMLS TUI gate.

**Gap 2 — Relation model.** MINERVA fine-tuned BioMistral-AUG-7B [35] (weights unreleased). We use Gemini 2.5 Flash-Lite zero-shot with 7-sample self-consistency [26] (full-consistency rate 0.896–0.916, a calibrated reliability measure, not a P/R claim). Closure path: fine-tune BiomedBERT on MicrobioRel (22 gut-microbiome relation types) or the Lever et al. 2023 labeled set [34].

**Gap 3 — UMLS normalization: closed and exceeded.** We match MINERVA’s CUI-linked deduplication (ScispaCy `en_core_sci_lg` [21], 2022 UMLS [20] KB; 96% / 83% CUI coverage on the initial / expanded corpora) and go beyond it by promoting the same CUI/TUI metadata to an *active acceptance gate*: a microbe failing to ground to a microbe-class TUI at similarity  $\geq 0.85$  is rejected pre-classification with an audited `drop_reason`. The TUI accept set `{T005,T007,T194,T204}` + non-microbe model-organism deny-list is documented and justified in Appendix A.

### 6. Limitations and Future Work

**Radiomic parameter abstraction.** The pipeline captures modality and body location but abstracts away bin width, voxel resampling, slice spacing, and ROI method—rarely reported in mined text. Two GLCM entropy values from different bin widths become one node. Future work: map to IBSI [18] feature-instance codes encoding extraction context.

**Demographic heterogeneity.** Body-composition metrics depend strongly on age, sex, and BMI; the graph aggregates across heterogeneous cohorts without a `PopulationContext` property. Future work: a cohort node or edge stratum enabling demographic filtering.

**Inter-scanner variability — a vision-track advantage.** By extracting *r*-values from heatmaps rather than absolute radiomic values, the vision track inherently normalizes across scanner hardware: a rank-order correlation is far more stable across sites than an absolute HU value, giving vision edges higher cross-site generalizability than value-anchored text associations.

**Observational only.** The graph encodes association, not causality. Future work: **Intervention** nodes (FMT, antibiotics, probiotics, diet) with typed **MODIFIES/REVERSES** edges, shifting toward a partially causal model—requiring longitudinal pre/post datasets

Table 2: Gap status relative to MINERVA.

| Gap                     | Sev. | Status                                                                                                                             |
|-------------------------|------|------------------------------------------------------------------------------------------------------------------------------------|
| NER (microbe)           | Med  | <b>Closed</b> via OpenMed-<br>NER-SpeciesDetect-<br>PubMed-335M (LIN-<br>NAEUS F1=0.9649);<br>TUI gate retained as<br>second guard |
| NER (disease)           | Low  | BC5CDR compet-<br>itive; OpenMed-<br>DiseaseDetect-<br>BioClinical identified<br>as A/B candidate                                  |
| Relation model          | Med  | Self-consistency 0.916;<br>fine-tune path docu-<br>mented                                                                          |
| UMLS norm.              | High | <b>Closed &amp; exceeded</b><br>(TUI-gated, audit-active)                                                                          |
| Text recall ceiling     | Med  | <b>Closed</b> (ModernBERT<br>dense retrieval)                                                                                      |
| Vision monocul-<br>ture | High | <b>Closed</b> (3 pre-verifier<br>gates + sign-check gate;<br>audit + reconciliation<br>9→7 edges)                                  |
| Measured P/R            | High | <b>Pending</b> Pass-2 (§4.5)                                                                                                       |

targeted in future harvest-lane design.

**Sparsity and track complementarity.** The three evidence layers are complementary by construction. The text track (phenotype→disease) is a broad associative network (74 edges, 30 diseases); the text track (microbe→feature) contributes 6 self-consistency-validated **CORRELATES\_WITH** edges; the vision track contributes 1 quantitatively verified edge plus the gating methodology. The microbe→feature edges are what close the 62 three-hop paths the intermediary-layer thesis requires; remaining **CORRELATES\_WITH** edges without downstream disease edges will close as text-track phenotype extraction expands to additional feature vocabularies. The graph is intentionally sparse and reviewable rather than dense and unaudited—this is a design stance, answering RQ3.

## 7. Conclusion

Inserting an explicit radiomics–body-composition intermediary between microbe and disease is feasible, mechanistically meaningful, and—given the five-gate acceptance model and a single provenance-stamped graph export—reproducible and reviewable. The vision track’s honest contribution is a hallucination-resistant verification methodology and one audited case study, not edge volume; the benchmark’s headline metrics are gated on a disciplined temporal window and reported as pending rather than estimated. The system’s defensibility comes from where it refuses to overclaim as much as from what it extracts.

**Reproducibility.** Code, tests, the reconciled `graph_export/` bundle with its provenance manifest, the Neo4j loader, and the read-only query layer are tracked in the project repository. The pre-reframing single-column proposal report is preserved unmodified at `docs/paper/proposal/`. Every numeric claim here is reproducible from the export manifest; gated metrics will be appended on the temporal window’s close.

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## A. Gate Threshold Derivations

Every numeric threshold in the five-gate model is derived from explicit reasoning rather than tuned-to-fit. We document them here so reviewers can challenge them directly.

### A.1 UMLS similarity floor (0.85)

The microbe-class TUI gate accepts a surface form only when the ScispaCy Entity Linker [21] returns a UMLS candidate [20] with cosine similarity  $\geq 0.85$ . Rationale: at 0.85, common typographical variants of well-known genera (“bacteroidetes” vs. “Bacteroidetes”) still ground correctly; below 0.80, clause-fragment noun phrases such as gut bacterial ClpB-like gene function (the surfaced Edge #5) ground as “Bacteria” (similarity 0.799) and would slip through. **Disconfirmation condition (pre-registered)**. If a live audit on the accepted microbe→feature edge set drops  $> 30\%$  of edges, the floor is over-strict and is recalibrated to 0.75. The 2026-05-07 live audit dropped 25% (2/8); the 0.85 floor is retained.

### A.2 TUI accept set {T005, T007, T194, T204}

The four-TUI accept set covers the UMLS Semantic Network’s microbe-relevant types: T007 (Bacterium), T194 (Archaeon), T204 (Eukaryote—broad enough to admit gut-fungal mycobiome entries), and T005 (Virus—admitted as part of a virome-extension future-proofing). T204 is broad enough to also admit model organisms (*Homo sapiens*, *Mus musculus*, *Rattus norvegicus*); to mitigate, an explicit non-microbe model-organism deny-list intercepts those CUIs before classification. Trade-off: a strict {T007}-only accept set would lose archaeal and fungal taxa entirely; the chosen breadth + targeted deny-list is the documented compromise.

### A.3 Default per-feature retrieval threshold $\tau = 0.62$

The dense-retrieval gate uses BioClinical-ModernBERT mean-pooled, L2-normalized embeddings indexed with FAISS [25] (NumPy cosine fallback at the current  $\sim 30$ k-sentence corpus). The pre-Pass-2 default  $\tau = 0.62$  is a defensible mid-range starting point chosen to (a) admit recall-probe-stratum paraphrases (“loss of muscle mass at L3”  $\leftrightarrow$  `skeletal_muscle_index`) while (b) excluding generic body tokens. **Disconfirmation condition.** If retrieval at  $\tau=0.62$  returns  $< 50$  candidates per common feature on the 1,016-paper corpus,  $\tau$  is under-tuned; per-feature calibration sweep follows once the Pass-2 labeled data is available.

### A.4 Vision range-sanity tolerance (0.05)

The range-sanity pre-verifier gate ( $|r_{\text{prop}}| \leq \max(|c_{\text{min}}|, |c_{\text{max}}|) + 0.05$ ) reads  $(c_{\text{min}}, c_{\text{max}})$  from the figure’s colorbar tick labels by a focused VLM call. The  $+0.05$  slack absorbs (a) numerical noise in the proposer’s  $r$ -value and (b) tick-label-rounding ambiguity (a colorbar labeled  $-1$  to  $+1$  may extend to  $\pm 1.05$ ). Without slack, a proposer value of  $\pm 1.00$  on a strict  $[-1, +1]$  scale would fail. With excess slack, the gate fails to catch the LFC ( $\pm 1.5$ ) vs. Pearson ( $\pm 1.0$ ) confusion that produced the retracted Firmicutes edge.

### A.5 Dual-verifier disagreement threshold (25%)

The dual-verifier (pixel HSV + independent VLM) routes XOR disagreements to a human-review queue. A disagreement rate above 25% indicates a verifier-design problem, not a content problem; the 2026-05-07 audit observed 46% disagreement on  $n=13$ , which prompted the introduction of the three pre-verifier gates (caption, colorbar-detect, range-sanity) that now catch the proposer hallucinations the dual gate alone shared a bbox with.

## B. Sample Size and Statistical Design

### B.1 Gold-set size (66 vs. 150 rows)

The benchmark sampler targeted 150 rows across five strata; the production candidate space yielded only 66. With a binary outcome and  $p \approx 0.7$ , 95% Wilson confidence intervals [24] are  $\pm 0.07$  at  $n=150$  vs.  $\pm 0.10$  at  $n=66$ . We accept the wider interval rather than artificially expanding the sampler to lower-quality strata; the Limitations section names this as a v1 honest constraint and sketches an expansion path (widen `entity_sentences` input to the broader 5,721-mention pool, with a re-tagging step).

### B.2 14-day intra-annotator window

Cohen’s  $\kappa$  [23] measured on an annotator’s two passes is biased upward when the second pass occurs too soon: short-term episodic memory of the first-pass label dominates the re-judgment. A 14-day window is the operational compromise between long enough to break se-

quential context and short enough that the annotator’s calibration to the schema does not drift. Pass-2 also starts from the UNLABELED file (not Pass-1) and uses randomized row order.

### B.3 Five-stratum sampling

Strata: `accepted_edge`, `gemini_rejected`, `vocab_excluded`, `recall_probe`, `random_co_occurrence`. Sampling is deterministic (seed 42) and reuses the production `_extract_candidates` function so the gold set evaluates against the actual production candidate space. Stratum weights are corpus-determined, not preset.

## C. Design Decision Rationale

### C.1 AND-consensus vs. majority / OR

The dual verifier accepts only on AND-consensus (both verifiers pass); XOR routes to human review. Majority/OR would silence verifier disagreement, which is exactly the signal that the audit trail needs.

### C.2 Direct-evidence-only policy

Within-paper bridge matches—microbe and feature mentioned separately in the same paper but never in the same correlation claim—are recorded to `bridge_hypotheses.jsonl` as audit-only hypotheses, never as graph edges. Reason: shared-paper context is necessary but not sufficient for causal/correlational assertion; admitting bridges silently inflates apparent edge count without raising evidence quality.

### C.3 ASSOCIATED\_WITH vs. CORRELATES\_WITH

The two-edge-type split is load-bearing. `CORRELATES_WITH` requires a quantitatively verified figure  $r$  or a 7/7 Gemini self-consistency text claim explicitly stating correlation between a microbe and a feature. `ASSOCIATED_WITH` marks text-co-mention feature $\rightarrow$ disease links without quantitative backing. Collapsing them would lose the precision signal that vision audits exist to preserve.

### C.4 ImageRef as a separate node type

Each verified vision edge spawns an `ImageRef` node carrying PMCID, figure ID, panel ID, topology, modality, and image path. This completes the chain `Disease $\leftarrow$ Feature $\rightarrow$ BodyLocation/ImagingModality $\rightarrow$ ImageRef`, making the evidence visually inspectable from the graph without re-running the pipeline.

### C.5 Retaining the *Prevotella nigrescens* vision edge despite a REVIEW verdict

The 2026-05-07 audit returned `final_verdict: REVIEW` (pixel\_inconclusive + vision\_pass) for this edge; we retained it after direct image inspection of PMC10605408\_g004 confirmed a real Spearman correlation heatmap with proper  $-1$  to  $+1$  colorbar and a  $r=0.95$  cell. The pixel-inconclusive



flicker was a single false negative on legend orientation, not a refutation of the underlying quantitative claim. The retention decision is recorded in `scripts/drop_failed_vision_edges.py` and the audit manifest.

## D. Stage A Reconciliation Algorithm

The reconciliation step deterministically composes (1) the assembled superset `artifacts/neo4j_relationships_microbe_expanded.csv` (191 rows / 9 CORRELATES\_WITH), (2) the UMLS audit drop-list `dropped_entities_audit.jsonl` (2 microbe→feature edges), and (3) the vision-audit retraction policy in `scripts/drop_failed_vision_edges.py` (1 edge: *Firmicutes*→Total fat %, PMC6178902\_g0006), onto a single 189-row base over 99 nodes. Both audits operated against different inputs and were never composed before this release; the manifest (`artifacts/graph_export/manifest.json`) records source-artifact vintages, the git commit SHA, pre/post counts, and every dropped row with its source-of-truth path. The manifest is the post-audit source of truth; every numeric claim in the body of this paper resolves against it.

**Reconciliation finding:** the pre-existing prose headline (“8 CORRELATES\_WITH: 1 vision + 7 text”) had applied only the vision retraction to the 9-edge superset and never composed in the separately recorded UMLS bacterioidetes drop. The coherent post-audit truth is 7 (1 vision + 6 text), 189 rows / 99 nodes, 62 three-hop paths.

## E. Body-Composition vs. Radiomics: Scope Distinction

The graph carries two distinct imaging-feature node classes that are sometimes conflated in casual discussion but should not be in a research-defensible artifact.

**RadiomicFeature** nodes are IBSI-standardized voxel-statistical features extracted from segmented image ROIs [16, 17, 18, 19]: GLCM (Gray-Level Co-occurrence Matrix), GLRLM, GLSZM, NGTDM, GLDM texture-matrix features, plus first-order intensity statistics, shape descriptors, and filter-derived features (LoG, wavelet, fractal). These match the formal pyradiomics scope.

**BodyCompositionFeature** nodes are clinical aggregate quantitative imaging biomarkers measured from CT/MRI/DXA, NOT from segmented-ROI texture analysis: skeletal muscle index ( $\text{cm}^2/\text{m}^2$  at L3), visceral adipose tissue ( $\text{cm}^2$  or  $\text{cm}^3$ ), sarcopenia [28], myosteatosis (mean muscle HU), psoas area, subcutaneous adipose tissue, body fat % (DXA), liver fat fraction (PDFF). These predate radiomics by decades and fall outside pyradiomics’ formal scope but inside the broader imaging-derived phenotype (IDP) / QIBA imaging-biomarker framework.

**Why both belong in this graph.** In the surviving literature corpus, 6 of 7 quantitatively verified microbe→feature edges target **BodyCompositionFeature**, not **RadiomicFeature**. The four established gut axes—muscle, liver, adipose, bone [27, 32, 33, 30]—all operate through body-composition imaging readouts, not IBSI textures. Excluding the **BodyCompositionFeature** node class would abandon ~95% of the literature evidence base. The architecture is the honest answer: two node classes, both imaging-derived, with the methodological precision posture (“radiomics-first”) and the literature coverage posture (“body-composition-dominant”) named explicitly rather than smuggled.

## F. Reproducibility Recipe

Exact commands to reproduce the graph from the repository (conda base on Apple Silicon; see `docs/RUN_CONTEXT.md` for runtime notes):

```
# 1. Reconcile the canonical export (idempotent)
conda run -n base python scripts/build_graph_export.py
# Outputs artifacts/graph_export/{nodes.csv, relationships.csv}
# import.cypher, manifest.json}
```

```
# 2. Regenerate the explorer JSONL from the canonical build
conda run -n base python scripts/build_explorer_data.py
# Backfills PMID/PMCID for vision edges; refuses to write if
# row-count drift vs manifest.json:post_total.
```

```
# 3. Bring up live Neo4j (operator-run; requires Docker)
export NEO4J_PASSWORD='choose-strong'
docker compose -f docker-compose.neo4j.yml up -d
conda run -n base python scripts/neo4j_load.py --dry-run
NEO4J_PASSWORD="$NEO4J_PASSWORD" \
conda run -n base python scripts/neo4j_load.py
```

```
# 4. Full test suite (351 checks; pins the load-bearing imports)
conda run -n base python -m pytest -q
```

Required environment variables for the optional paid-API paths: **GEMINI\_API\_KEY** (relation extraction; vision verifier), **NEO4J\_PASSWORD** (live load only). UMLS scripts must be run from **Terminal.app** (not VSCode terminal) due to a ScispaCy + UMLS KB process-isolation requirement (`docs/NEXT_STEPS.md` “Runtime Notes”).